

Research Summary: Impact of Generative AI on Software Development

1. Introduction

Generative Artificial Intelligence (GenAI) is changing the way software is developed. AI tools can assist developers with writing code, debugging, documentation, testing, code explanation and other development activities. As these tools become more capable, the role of software developers is also changing from performing routine coding tasks toward problem-solving, system design, reviewing AI-generated output and making technical decisions.

A recent World Economic Forum article published on **17 August 2026**, titled “**How AI is reshaping the path from junior to senior developer**,” discusses how AI is changing software-development work and the skills developers need in the AI era.

2. Technology Discussed

The main technology discussed is **Artificial Intelligence, particularly Generative AI** used in software development.

Generative AI can produce code and assist developers with tasks that previously required manual effort. This allows developers to complete some routine tasks faster. AI can also support developers by generating suggestions, explaining code, helping identify problems and assisting with technical workflows.

However, AI-generated output still needs to be reviewed by developers. Human understanding is important because developers need to determine whether the generated solution is correct, secure, suitable for the business requirement and maintainable.

The article highlights that as AI takes over more routine development work, developers are increasingly required to understand **architecture, domain knowledge, technical judgment and AI-system orchestration**.

3. Business Impact

Generative AI can provide businesses with several potential benefits:

- Higher productivity: Developers can complete certain repetitive tasks faster.
- Faster development: AI assistance can reduce the time required for coding, documentation and debugging.
- Automation: Routine development activities can increasingly be supported or automated.
- Changing workforce requirements: Companies need developers who can effectively work with AI tools.
- New training requirements: Organizations need to provide opportunities for developers to gain practical experience while using AI responsibly.

The business impact is not limited to productivity. The World Economic Forum article highlights a potential challenge: if AI performs many of the routine tasks traditionally given to junior developers, junior employees may receive fewer opportunities to gain practical experience. The article reports that **54% of senior developers surveyed agreed that AI is making the junior role less relevant**.

Therefore, organizations need to balance AI-driven productivity with proper training and career-development opportunities.

4. Impact on Software Developers

The role of developers is gradually shifting.

Earlier, junior developers often gained experience by performing smaller coding tasks, fixing bugs and working on routine development activities. With AI handling some of these tasks, developers may need to focus earlier on higher-value activities.

The article identifies areas such as critical thinking, problem-solving, fundamental technical knowledge, problem decomposition, domain knowledge, software architecture, AI-assisted development, and reviewing and auditing AI-generated output.

The article reports that among senior developers surveyed, **critical thinking was the most frequently selected important skill for junior developers, at 25%, compared with 18% for AI literacy.**

This shows that learning AI tools alone is not enough. Developers also need strong fundamentals and the ability to evaluate the results produced by AI.

5. Skills Developers Should Learn

To remain relevant in an AI-driven software industry, developers should build a combination of traditional software-development skills and AI-related skills.

- **Technical Skills:** Programming fundamentals, data structures and algorithms, database management, API development, Git and GitHub, testing and debugging, cloud and deployment, system design.
- **AI Skills:** Generative AI fundamentals, Large Language Models (LLMs), prompt engineering, AI API integration, RAG and embeddings, AI agents, AI-assisted coding, evaluating AI-generated output.
- **Human and Professional Skills:** Critical thinking, problem-solving, communication, analytical thinking, domain knowledge, decision-making, continuous learning.

The article particularly emphasizes **real-world project experience**, with 49% of junior developers surveyed selecting it as the area where training should focus more.

6. Key Learning Points

- 1 Generative AI is changing the software-development workflow.
- 2 AI can automate many routine coding and development activities.
- 3 Developers still need strong programming fundamentals and problem-solving skills.
- 4 Critical thinking is important when reviewing and validating AI-generated solutions.
- 5 Developers should learn how to use AI as a development tool rather than depend on it blindly.
- 6 AI is increasing the importance of architecture, domain knowledge and technical judgment.
- 7 Organizations need to provide real-world learning opportunities so developers can continue developing practical skills.

7. Conclusion

Generative AI is not simply changing how code is written; it is changing the responsibilities and skills expected from software developers. Developers who combine strong technical fundamentals with AI knowledge, critical thinking and practical experience can adapt to this changing environment.

For businesses, the challenge is to use AI to improve productivity while continuing to develop the next generation of software professionals. The future of software development is therefore likely to involve greater collaboration between developers and AI systems, with humans responsible for understanding requirements, making decisions, validating results and managing complex technical systems.

8. Article Reference

Article: How AI is reshaping the path from junior to senior developer

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